MANCHESTER
1824
The University of Manchester

Department of Electrical and
Electronic Engineering

Embedded Systems Project

DESIGN REPORT #1

Title: Motor Analysis and Gearbox Selection

Yiwen Yao (Andrew) 11009023

Tutor: Wuqiang Yang

Contents

1. Introduction

The successful development of the autonomous buggy hinges on the seamless integration of software, firmware, and hardware. It is paramount that through deliberated thought, components are selected from each sector which will harmonise. This report delves into the hardware aspect, in particular the DC motor and gearboxes.

The motors and sensors can be considered the foundation of the project, collecting and transmitting data for the software's interpretation. One of the aims of the project is to optimize data accuracy in collection and ensure smooth communication between the hardware and software; however, the project is subject to various constraints that may challenge these aspirations. One of these constraints is the requirement to use the provided motor and select from three gearboxes, necessitating an in-depth understanding of their specifications and requiring alterations to other parts of the project to accommodate for their shortcomings. In addition to this, the directive to use only the batteries provided in the given packs, excluding LiPo batteries, places limitations on the power source. This necessitates a careful consideration of the power requirements of the motors and sensors, aligning them with the capabilities of the designated battery packs. A comprehensive analysis of the motors and gear boxes are required to create an efficient solution that not only adheres to the specified constraints but also maximizes the potential of the provided motors and gearboxes.

Since the design of motors used is limited, there is also a constraint on the torque they provide. In this case, the given motor does not provide adequate torque to get the buggy up the slope, this constraint is why gearboxes will be essential in the project. One of gearboxes fundamental functions is torque multiplication, since the torque produced by a motor is related to its speed through the power equation:

$$\text{Power} = \text{Torque} \times \text{Speed}$$

However, gearboxes can add weight and occupy large spaces on the vehicle, two factors that must be considered when creating the chassis.

A gearbox is not the only way to alter the speed or torque of a system. A motor's speed and torque can be dictated by electrical signals in the form of voltage and current. Voltage signals can directly affect the speed of the motor: By increasing the voltage supplied, the initial effect is to increase the voltage across the armature. The higher voltage across the armature tends to increase the armature current. As the armature current increases, it generates more torque, causing the motor to accelerate. However, as the motor speeds up, the back EMF also increases, opposing the increase in armature current. Eventually, a new balance is reached where the back EMF equals the applied voltage, and the motor stabilizes at a higher speed. A similar process is followed with current signals. This means that both voltage and current can be used as signals to control the motor speed. However, the goal of this project is for the buggy to be autonomous, to do this, a microcontroller is required. The microcontroller will be encoded with software that responds to external stimuli by interfacing with the motor drive and coordinating its movement.

The microcontroller can interface with the motor drive board using a PWM signal. Pulse width modulation is a method of creating an approximated analogue signal from a digital output pin, it varies the duty cycle of a square wave signal to control the average voltage supplied to a motor, thus being able to control the speed and direction of said motor via the drive board. PWM allows us to convert the square signals from the microcontroller to analogue signals which will go to the motor.

The aim of this report is to select a gearbox which will provide maximum speed for traversing the given track, whilst also providing enough torque to ascend the largest slopes.

2. Motor characterisation

2.1. Understanding the characteristics of the motor is integral to the overall success of the project. Voltage and current supplied will determine its rotational speed and torque, which will determine the buggy's speed and ability to ascend slopes. This section serves as the foundation upon which the entire project relies. It is closely intertwined with the report's aim and objectives, since the results from the lab allowed us to understand how a motor can be characterised by mechanical and electrical parameters that are interrelated like the motor torque, back emf and rotational speed ω.

2.2. The motor is expected to operate within specific constraints. The maximum voltage and current the drive board will supply to each motor is 5V and 1.4A, to avoid overheating. This limits the options for the gearbox selection, as sufficient torque must be generated by the motor and gearbox combination to push the buggy up an 18-degree incline slope while receiving a current supply of less than 1.4 A.

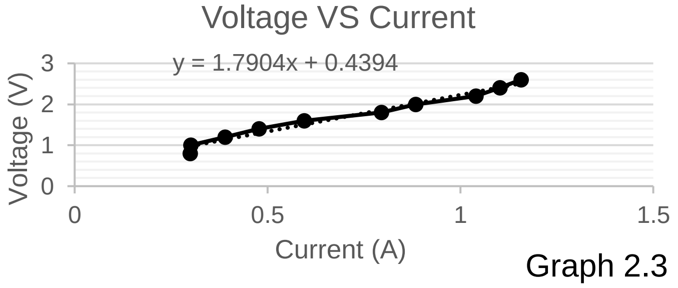
2.3. A series of experiments were conducted to gather test results and derive values for the motor. The process began with the measurement of the resistance of the PSU leads to know the voltage dropped across them when applying current to the motor.

The resistances of the leads were found to be 0.0573Ω and 0.0553Ω, from the gradient of the graphs.
Next, the motor voltage was measured by connecting the voltage measurement unit directly to the motor while applying a current of 0.118 A. The measured motor voltage registered at 9.996 V, and the motor resistance was subsequently determined as the ratio of the motor voltage to the current, yielding a value of 84.71186 Ohms.

Then for task 8.3 Armature Resistance Measurement and brush voltage. The motor voltage was set to about 1 V and made sure the over-current protection limit is 1.7 A, then loaded the shaft to stall the motor and tried 10 different values of motor voltage, up to the maximum current (1.4 A) and these are the values:

Current (A)	Voltage (V)
0.299	0.8
0.301	1
0.39	1.2
0.478	1.4
0.595	1.6
0.795	1.8
0.884	2
1.04	2.2
1.102	2.4
1.157	2.6

Then the Graph 2.3 is determined the motor resistance R and the brush voltage V_b:

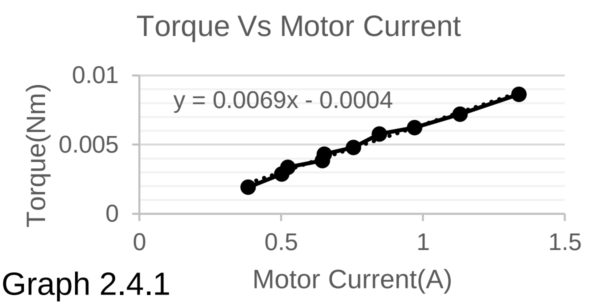


Graph 2.3

The motor resistance R is equal to the slope of the graph so R = 1.7904 Ohms and the brush voltage V_b is equal to the y-intercept of this graph so V_b = 0.4394 V

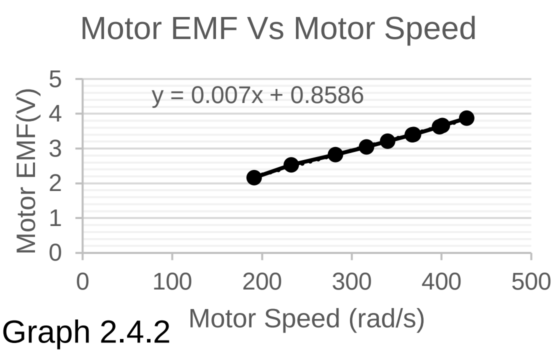
2.4. Using Vernier callipers, the diameter of the disc shaft was measured to be 0.0096m. After connecting a piece of string to 2 force gauges and connecting it the motor to simulate torque using string tension, the motor was supplied with a voltage of 5 V. Values of motor rotation per minute was recorded using an optical tachometer, and current was recorded using the digital power supply.

After obtaining 10 sets of values for the force gauges, torque was calculated using the equation T = (F1-F2) (d/2). Then, with recorded values of torque and current, the



Graph 2.4.1

Graph 2.4.1 of torque and current was plotted, giving the gradient K_T as 0.0069 and $T_{friction}$ = 0.0004Nm

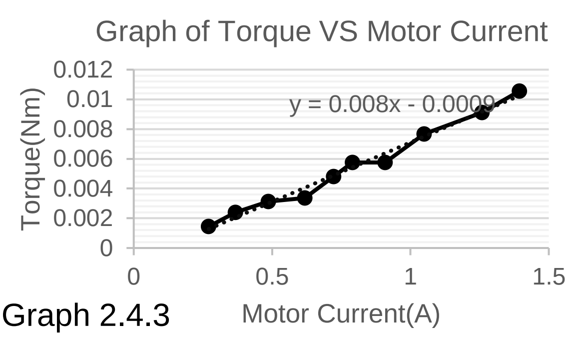


Graph 2.4.2

To estimate Ke, the equation is

Ke = back emf / motor speed was used. By plotting motor emf against motor speed in (rad/s) can be shown in Graph 2.4.2, the gradient of the graph gives Ke = 0.0070

Using the same setup, Graph 2.4.3 can be shown that K_T was estimated again but at stall (high torque).



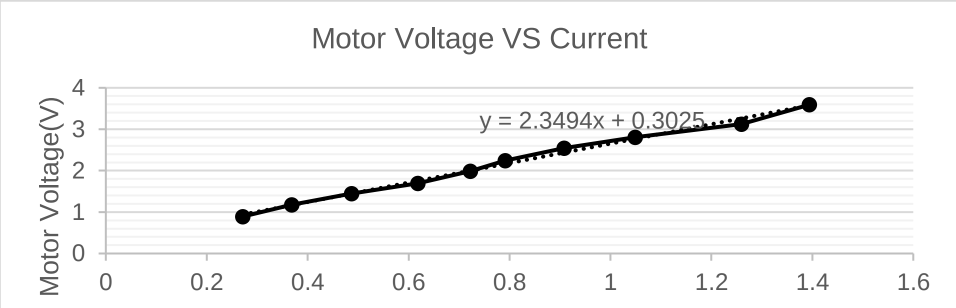
Graph 2.4.3

And K_T is equal to the gradient of Graph 2.4.4 so K_T = 0.0080, then Plotted motor voltage against the current.

And got R (resistance) from the gradient of the plotting so R = 2.3494 Ohms. and then the brush voltage V_b is the y-interception of this plotting so V_b = 0.3025 V.

2.5. However, it is also important to note that there are slight uncertainties in the results. Firstly, the digital multimeter used during the lab gave results up to 3 d.p., leading to a 0.0005 V absolute uncertainty in each result. Another cause of uncertainty is due to voltage drop across the PSU leads used to conduct the experiment. It was recorded that each PSU lead had a resistance value of 0.058Ω and 0.061Ω respectively. This value was not factored into the calculations, leading to inaccuracies in the lab results.

The Graph 2.4.4 can be shown that below:

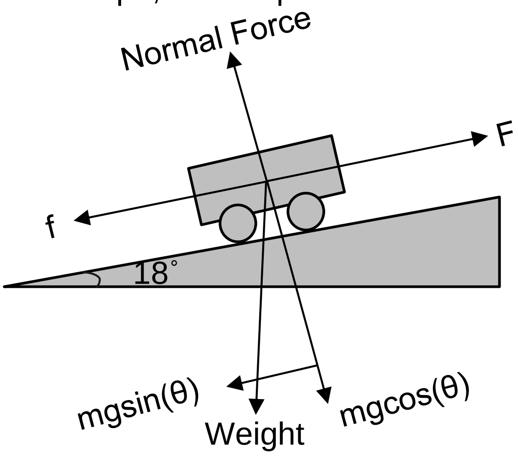


Graph 2.4.4

3. Load measurements

3.1. This section is for driving the buggies with different weights at a constant speed on slopes at different angles in which the force applied to the buggy to move up the slope will be measured. Then a comparison was made for the torque applied in the different gearbox ratios to choose the best gearbox. In this section, the assumption is ignoring the air friction.

3.2. When the buggy up the slope, the Graph 3.2.1 is Force Analysis Diagram,



Graph 3.2.1

where f is friction, F is tension, m is total mass of buggy, g is a constant and is 9.81 approximately. The buggy had a mass m = 1.246kg, at a slope θ = 18˚, and the force required was F = 4.905N to move it with a constant speed.

For this section, the group was tasked adding the weight above the buggy and driving the buggy up to the slope with constant speed, then reading the tension by using force gauge.

To calculate the coefficient of friction, the friction must be known,

$$f = F - mg \times \sin(\theta)$$

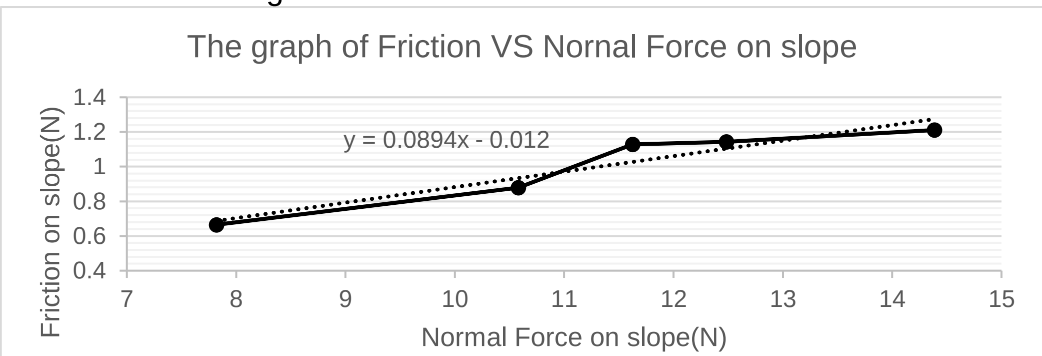
$$f = 4.905\text{N} - 1.246\text{kg} \times 9.81\text{Nkg}^{-1} \times \sin(18^\circ) = 1.128\text{N}$$

And then using $f = \mu F_N$ where μ is the coefficient of friction and F_N is Normal Force.

$$\mu = \frac{1.128\text{N}}{1.246\text{kg x } 9.81\text{N}kg^{-1} \cos(18^\circ)}$$

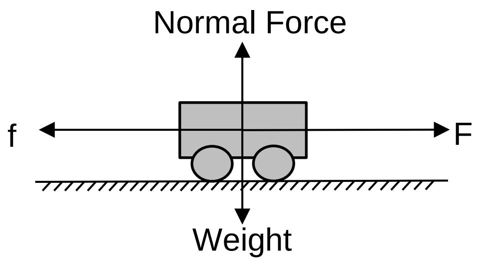
$$\mu = 0.097$$

The relationship between Friction force and Normal force is shown that Graph 3.2.2, it is found that the average coefficient of friction is 0.0894.



Graph 3.2.2

3.3. When the buggy is driven on the flat, the Graph 3.3.1 is Force Analysis Diagram,



Graph 3.3.1

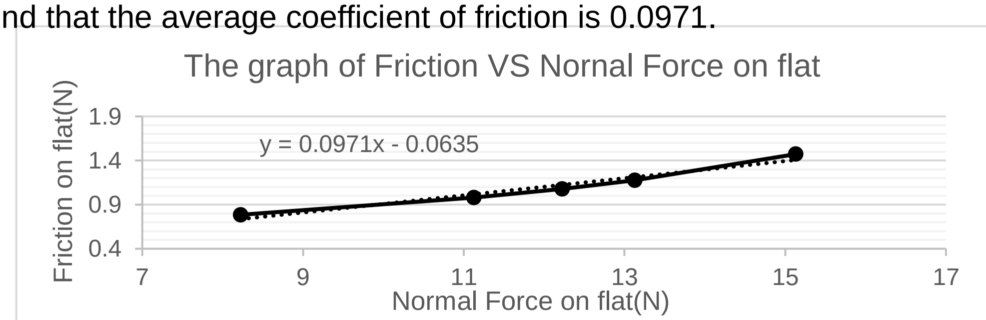
where f is friction, F is tension, Weight is mg, Normal Force is equal to Weight. The initial measurements where m = 1.246kg, F = 1.0791N,

For this experiment, weights were added above the buggy, and the buggy was driven with constant speed on the flat, then reading the tension of buggy by using force gauge.

Due to constant speed, f is equal to F. Using $f = \mu F_N$ where μ is the coefficient of friction and F_N is Normal Force.

$$\mu = \frac{1.0791\mathrm{N}}{12.223\mathrm{N}} = 0.0883$$

The relationship between Friction force and Normal force is shown that Graph 3.3.2, it is found that the average coefficient of friction is 0.0971.



Graph 3.3.2

For 2 and 3, the coefficient of friction is close to equivalent. The difference between these due to the random error during the laboratory.

3.4. For the torque at wheel shifts, the radius of wheel is necessary. In the experiment, the vernier callipers were used to measure the diameter of wheel and it is about 77.1mm.

$$r = radius = \frac{diameter}{2} = 38.55\text{mm}$$

To calculate the torque, $T = Fr$, where F is tension, r is radius of wheel.

For this calculation, assuming that the weight of buggy is 1.246kg.

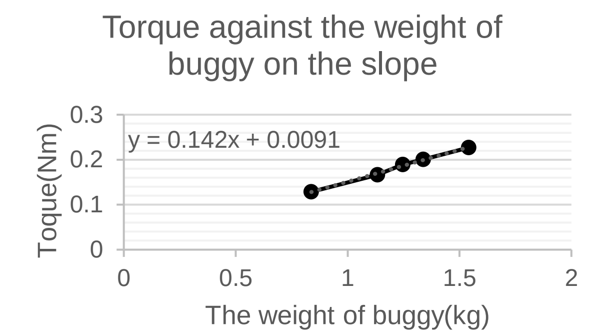
When the buggy is driven on the slope,

$$F = 4.905\text{N},\ T = 4.905\text{N x } 38.55\text{mm} = 0.189\text{Nm}$$

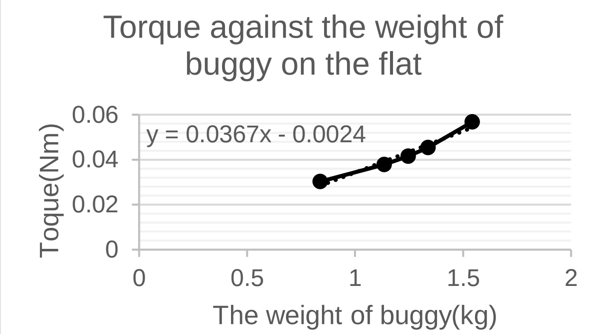
When the buggy is driven on the flat,

$$F = 1.4715\text{N},\ T = 1.4715\text{N x } 38.55\text{mm} = 0.0416\text{Nm}$$

For some different values of weight, the Graph 3.4.1 and Graph 3.4.2 can show that,



Graph 3.4.1



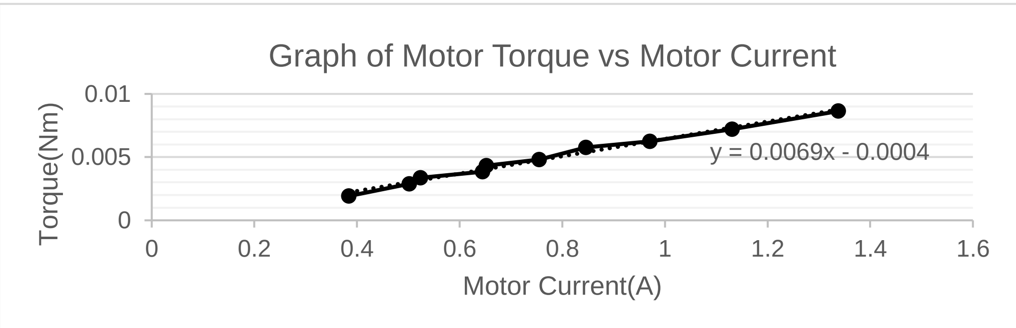
Graph 3.4.2

Hance, the torque is proportional to the weight of buggy.

4. Gear ratio selection

4.1. The main aim of this section is to select the appropriate gear ratio to pair with the given motor, so that they can generate enough torque to push the buggy up the maximum anticipated incline (18 degrees). The typical efficiency of a well-constructed gearbox is 85% per gear stage.

4.2. The force required to get the buggy up the max incline would be given by "μmgcos(θ) + mgsin(θ)", as shown in Graph 3.2.1. The coefficient of friction on the given surface, μ , is found to be 0.097 in calculations from the Load Measurement section. Assuming m = 1.246 kg, g = 9.81, θ = 18 degrees, the expression is that μmgcos(θ) + mgsin(θ) gives us a force of 4.904 N required to move the buggy up the slope. With T = Fr, F = 4.904N and radius of the wheel was measured to be 0.03855m. T = 4.904N x 0.03855m = 0.1891 Nm. With 2 wheels, the minimum torque needed on each wheel would be 0.0945 Nm.



Graph 4.2

From Graph 4.2, K_T can be found by the gradient f the graph, K_T = 0.0069

With the equation T = K_T I, the minimum current needed required to generate enough force for an 18-degree incline can be calculated using I = 0.0945/0.0069 = 13.7 A.

4.3. The maximum current supplied by the drive board to the motor is assumed to be 1.4 A. Using the equation T = K_TI = 0.0069 x 1.4 = 0.00966 Nm, the maximum Torque will be 0.00966 assuming 100% efficiency.

4.4. The following graph is the three different Gears that be selected:

Selection Number	Pinion Gear (motor shaft only)	Intermediate Gear Two common gears fixed on one shaft	Gear on the final drive
1	16 tooth	48/12 press fit gears (orange*)	48 tooth
2	16 tooth	50/10 press fit gears (orange)	48 tooth
3	16 tooth	50/10 press fit gears (red)	60 tooth

Out of the 3 available gear ratios, assuming an 85% gearbox efficiency, Gear 1 provides a gear ratio of 10.2, Gear 2 provides a gear ratio of 12.75 and Gear 3 provides a gear ratio of 15.94. After factoring in the gear ratios, it was found that the current required to generate enough torque for a 1.246 kg buggy to go up an 18-degree slope for each gear ratio to be 13.7A/10.2=1.343A, 13.7A/12.75=1.075A, and 15.72A/15.94=0.860A respectively. Assuming the maximum current supplied by the drive board to the motor to be 1.4 A, Gear Ratio 1 is a risky choice, as the weight of the buggy is not yet determined, so the assumption of 1.246 kg is unreliable. While Gear Ratio 3 also guarantees enough torque, this gear-motor combination generates excessive torque for an 18-degree incline, which would not be wise as torque is inversely proportional to speed. Therefore, Gear Ratio 2 shall be chosen, as it guarantees sufficient torque at a current supply of less than 1.4 A without sacrificing too much speed.

4.5. PCD = Number of teeth x MOD.

Centre distance between pinion gear and intermediate gear

= (PCD(A) + PCD(B_1))/2 + 0.1 = (8 + 25)/2 + 0.1 = 16.6 mm

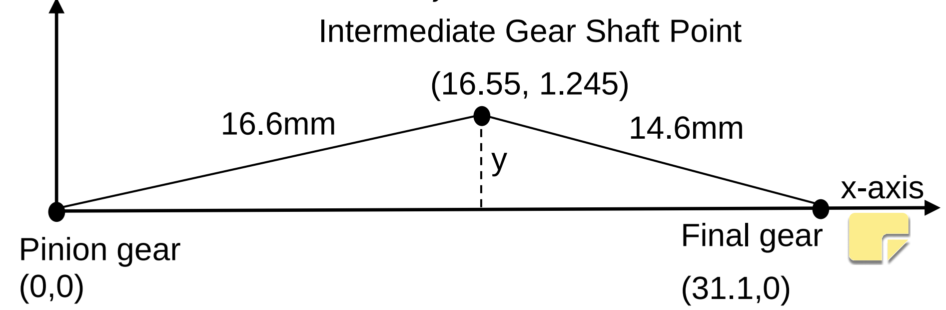
Centre distance between intermediate gear and gear on the final drive

= (PCD(B_2) + PCD(C))/2 + 0.1 = (5 + 24)/2 + 0.1 = 14.6 mm

As the total distance between the pinion gear and final gear is fixed at 31.1 mm, the coordinate of the intermediate shaft can be found using the equation:

$$\sqrt{16.6^2 - y^2} + \sqrt{14.6^2 - y^2} = 31.1$$

$$y = 1.245$$

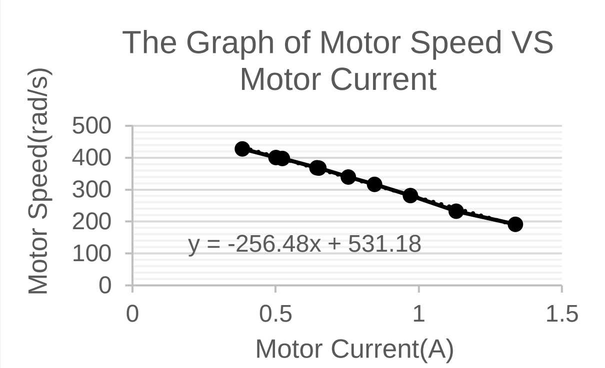


4.6. On flat ground, the force required to push the buggy assuming a mass of 1.246

kg is given by F = μmg = 0.0883 x 1.246 x 9.81 = 1.079 N. Using T = Fr, the torque required to generate 1.0793 N, T = 1.0793 x 0.0386 = 0.0419 Nm. Each wheel must generate 0.0210 Nm of torque. With a gear ratio of 12.75 and the equation I = T/K_T, the minimum current required for the buggy to move on flat ground I = 0.238 A. From the motor lab results, a current of 0.238 A would result in a motor speed of 470.17 rad/s. This can be shown in Graph 4.6 below.

Then, using the formula $\omega_3 = \omega_1 \times N_1/N3$ = 470.17 x 16/48 = 156.7rad/s motor speed. Lastly, with v = ωr = 156.7 x 0.03855 = 6.04, so the max speed of the buggy on flat ground = 6.04 m/s.

Using the same method, the max speed of the buggy on a slope can be found by substituting the force as 4.904 N instead of 1.079 N. The torque required from each wheel was found to be 0.0945 Nm, and from that the minimum current supplied to the motor was found to be 1.074 A. From the lab results, the motor speed would be 261.1 rad/s, which would mean a max speed of 3.36 m/s on slope.



Graph 4.6

5. Summary

To give a conclusive analysis on the results of this report, the main objective of the report must be recalled, which is to pick out a gearbox which will optimize the movement of the buggy on the track. This report also helps us enhance the precision of data gathering while maintaining a seamless communication between the physical components and the software system, which will help us create an effective and streamlined motor-powered mechanism for the buggy.

Through experiments done in the Motors Lab, sufficient data was collected to perform calculations deriving the motor constants such as K_T ([illegible]69) and K_E (0.0070), as well as finding out the maximum voltage and current which will be applied in the motor drive board (5V and 1.4A).

In the end, gearbox 2 turned out to be the appropriate gearbox because assuming the weight of the buggy m = 1.246 kg, gearbox 2 produces enough torque to drive the buggy up an incline of 18 degrees with a current supply of 1.075 A, as calculated above. This leaves room for error on the assumption of m = 1.246 kg, which is not yet determined.

6. References

[1] ESP Technical Handbook, The University of Manchester (2023).Available: https://online.manchester.ac.uk/webapps/blackboard/execute/content/file?cmd=view&content_id=_15044517_1&course_id=_78308_1&launch_in_new=true

[2] ESP Procedures Handbook, The University of Manchester (2023).Available: https://online.manchester.ac.uk/webapps/blackboard/execute/content/file?cmd=view&content_id=_15044516_1&course_id=_78308_1&launch_in_new=true